architecture, which may include septations, calcifications, wall thickening, or enhancement patterns that raise concern for malignancy.

On imaging, simple cysts demonstrate characteristic features including homogeneous fluid attenuation, thin imperceptible walls, and absence of contrast enhancement, allowing confident diagnosis without further evaluation. Complex cysts, in contrast, exhibit heterogeneous appearances that require careful characterization to assess malignant potential. The primary imaging modality for cyst evaluation is computed tomography (CT) with contrast enhancement, though magnetic resonance imaging (MRI) provides superior soft tissue characterization in select cases and offers advantages in differentiating benign from malignant lesions through advanced sequences [7].

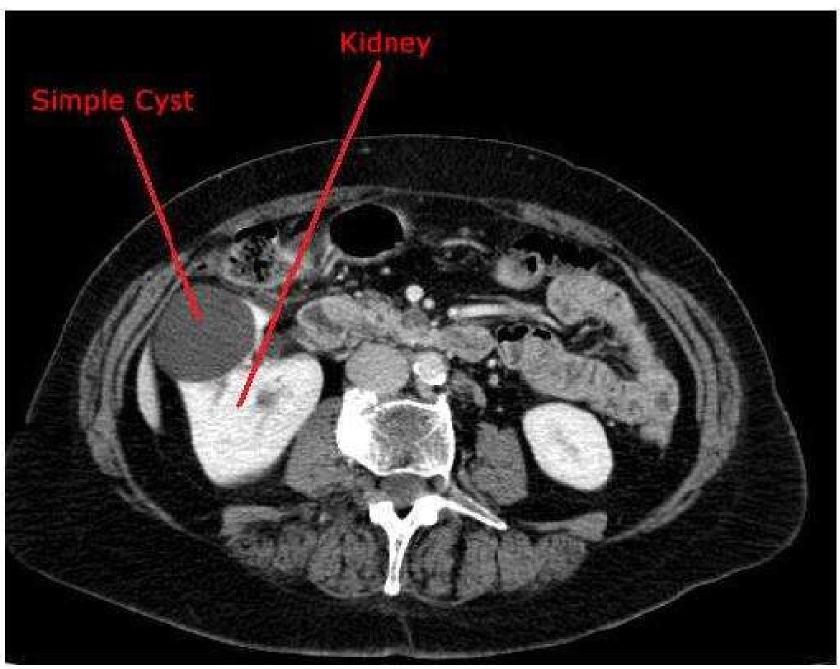

Figure 1. Simple Cyst on CT scan. Reprinted from [23].

The clinical significance of accurate cyst characterization lies in distinguishing lesions that require surveillance or intervention from those that can be safely ignored,

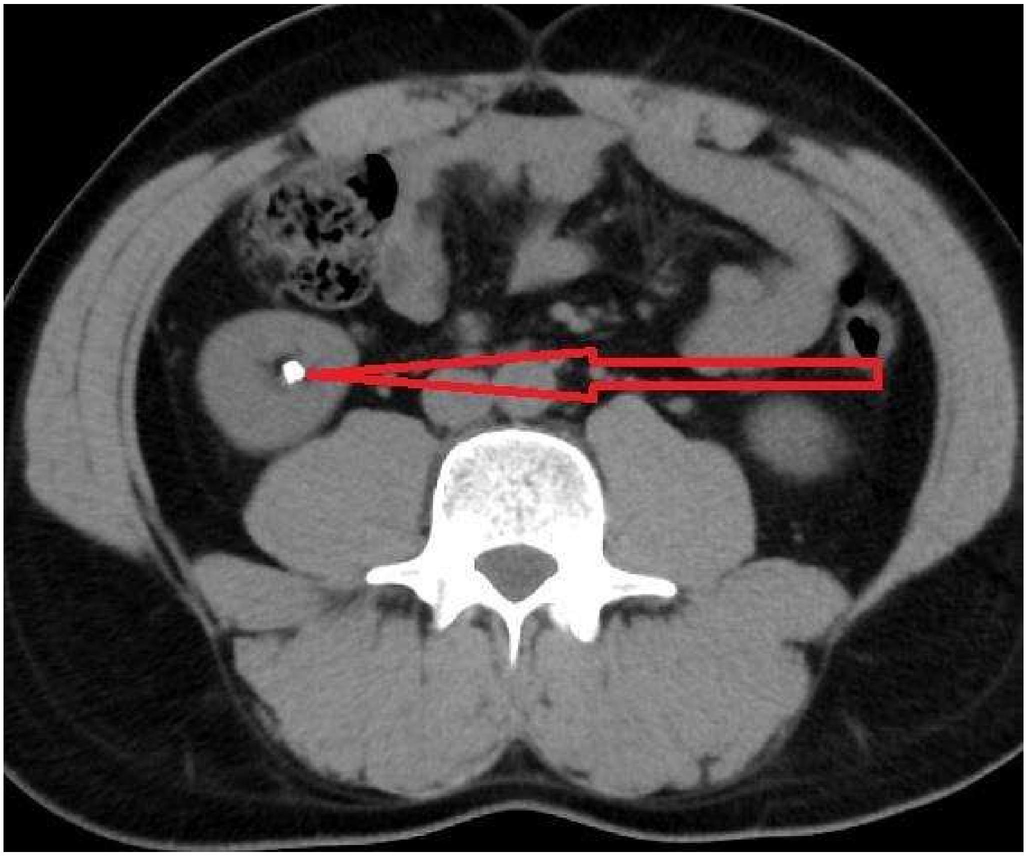

Figure 2. Stone on CT scan. Reprinted from [24].

Beyond acute diagnosis, imaging plays a crucial role in monitoring stone burden, detecting new stone formation, and evaluating treatment response. The accurate detection and measurement of kidney stones is clinically essential because stone characteristics directly determine whether patients require intervention or can be managed conservatively, and precise quantification of stone burden informs prognosis regarding recurrence risk and potential for progressive kidney damage.

2.1.4 Tumors: Clinical Presentation and Imaging Characteristics

Renal tumors encompass a diverse group of neoplastic lesions that may arise from various cellular origins within the kidney, with the majority detected incidentally during abdominal imaging performed for unrelated indications.

The dramatic increase in incidental tumor detection has fundamentally changed the epidemiology of renal neoplasms, with approximately 60-70% of cases now discovered asymptomatically compared to historical presentation patterns dominated by the classic triad of hematuria, flank pain, and palpable mass, and recent series from 2020-2022 confirm less than 5% of patients present with these classic symptoms [5].

Imaging plays a pivotal role in tumor detection and characterization, with contrast-enhanced CT and MRI serving as primary modalities for evaluating lesion size, enhancement patterns, and anatomical relationships. Renal tumors demonstrate heterogeneous imaging characteristics, with solid enhancement following contrast administration serving as the key discriminating feature from cystic lesions (see Figure 3) [7], [8].

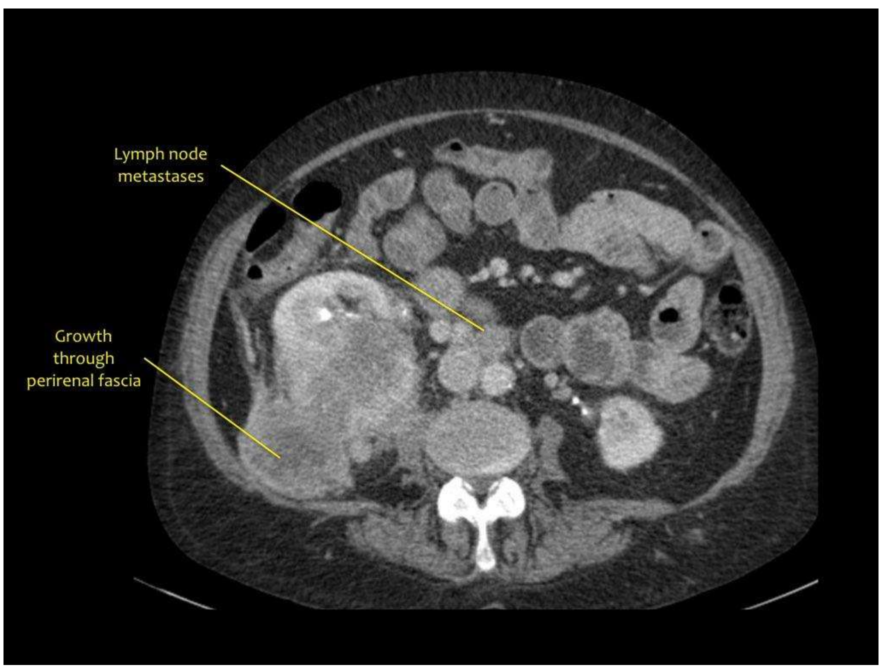

Figure 3. Solid Renal Tumor on CT scan. Reprinted from [25].

However, significant overlap exists in the imaging appearances of various tumor types, and no single imaging feature reliably distinguishes between different pathological

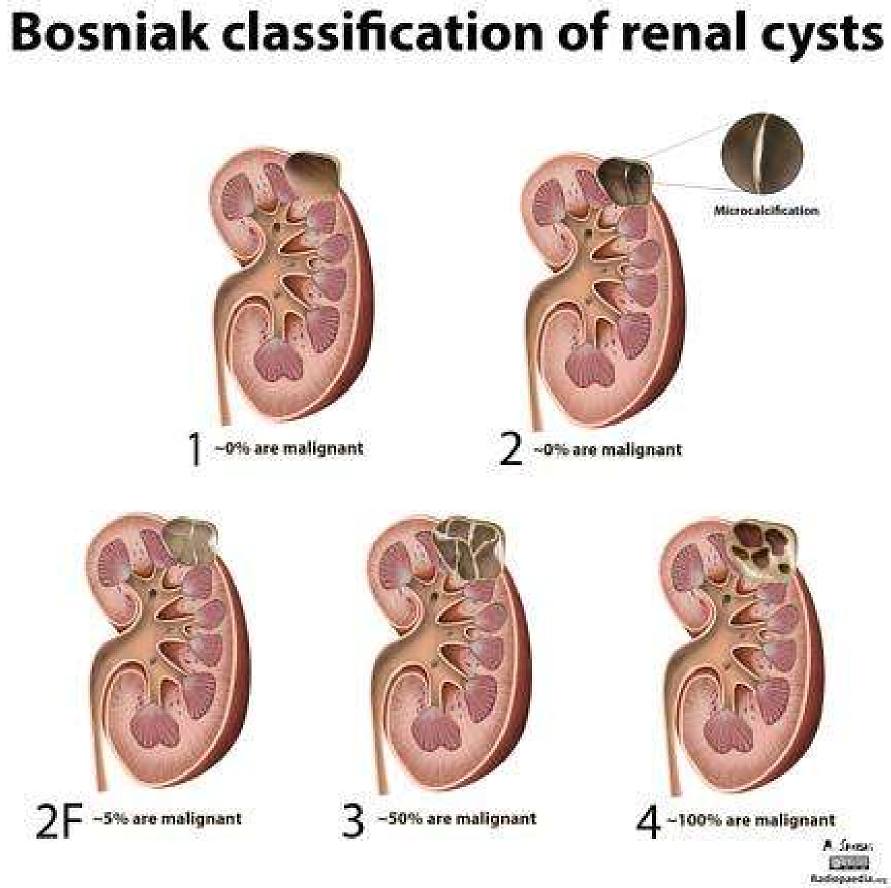

Figure 4. Bosniak Classification System. Reprinted from [27].

This system stratifies cysts into categories ranging from benign simple cysts requiring no follow-up to complex lesions with high malignant potential requiring surgical intervention, with intermediate categories necessitating surveillance imaging. While widely adopted, recent studies continue to demonstrate that the Bosniak system exhibits inter-reader variability in category assignment, with absolute discrepancy rates ranging from 6% to 75% and particularly pronounced disagreement for Bosniak II, IIF, and III masses, leading to inconsistent management recommendations [28], [29]. For renal masses, the R.E.N.A.L. nephrometry scoring system quantifies tumor complexity based on radius, exophytic properties, nearness to collecting system, anterior/posterior location, and location relative to polar lines (depicted in Figure 5), primarily serving to predict surgical difficulty and guide treatment selection.

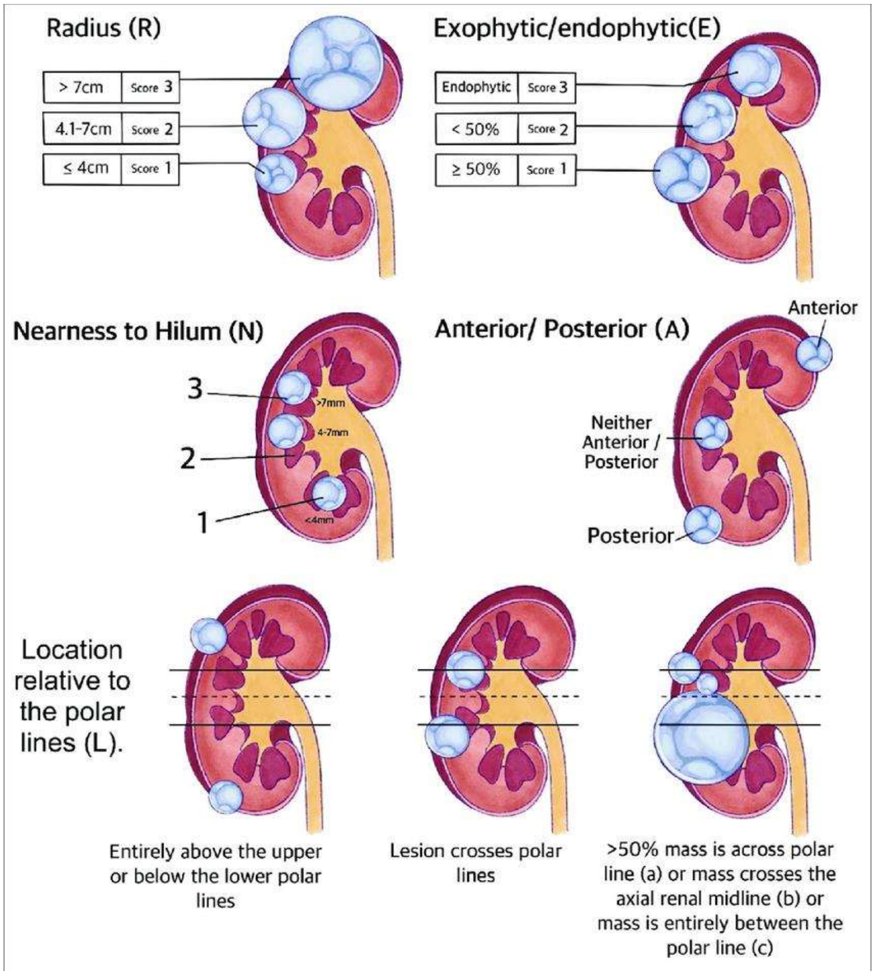

Figure 5. R.E.N.A.L. Nephrometry Score. Reprinted from [30].

These classification frameworks, while valuable, rely on subjective human interpretation of imaging features and are susceptible to inter-observer variability, particularly in borderline or complex cases. The inherent subjectivity in applying these systems, combined with the increasing volume of detected abnormalities requiring characterization, creates inefficiencies in radiology workflows and potential inconsistencies in patient management, highlighting the need for more objective and reproducible methods of abnormality characterization.

coordinates and class labels for detected objects [33]. In contrast, as illustrated in Fig. 1, segmentation extends beyond simple localization to delineate precise pixel-level boundaries of anatomical structures or pathological findings, enabling detailed morphological analysis and volumetric measurements.

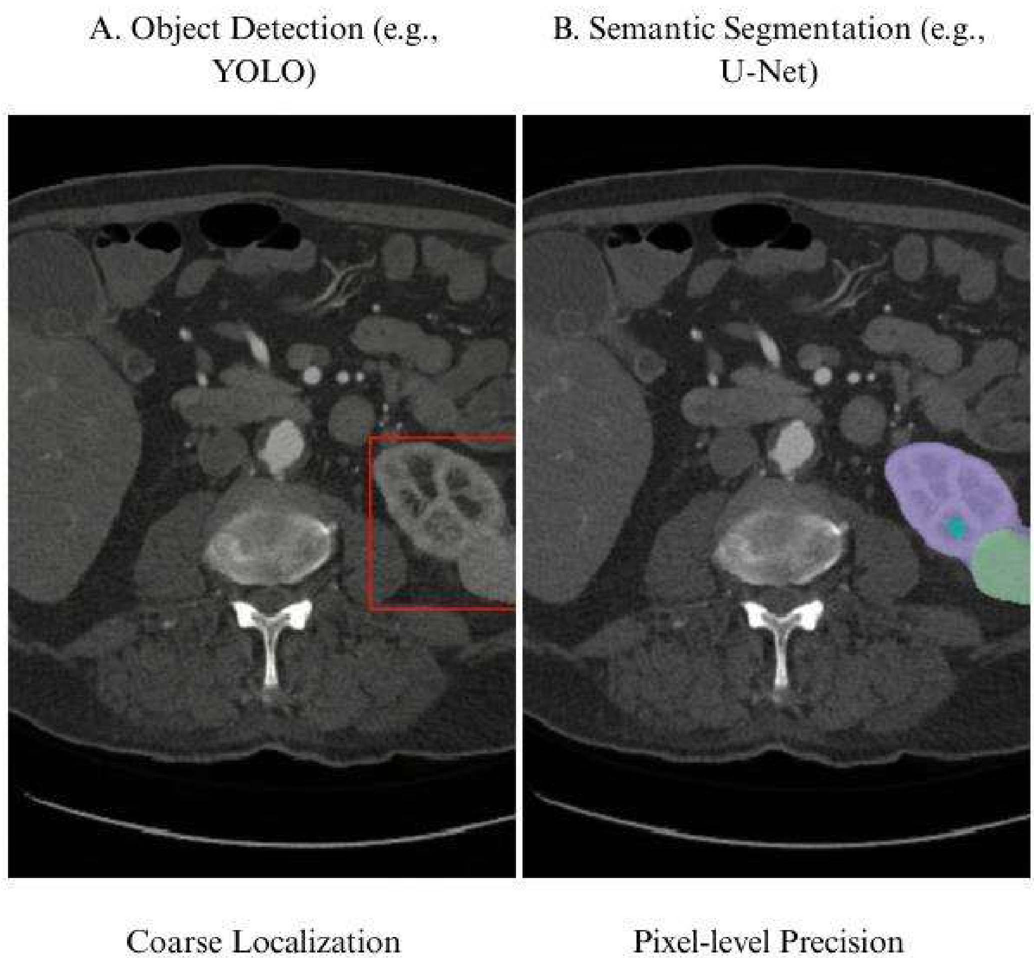

Figure 6. Comparison of Object Detection versus Semantic Segmentation in Medical Imaging. Adapted from [34].

These methodologies operate synergistically in urological imaging workflows, where detection algorithms first identify suspicious regions that may contain kidney abnormalities, subsequently guiding segmentation algorithms to perform detailed

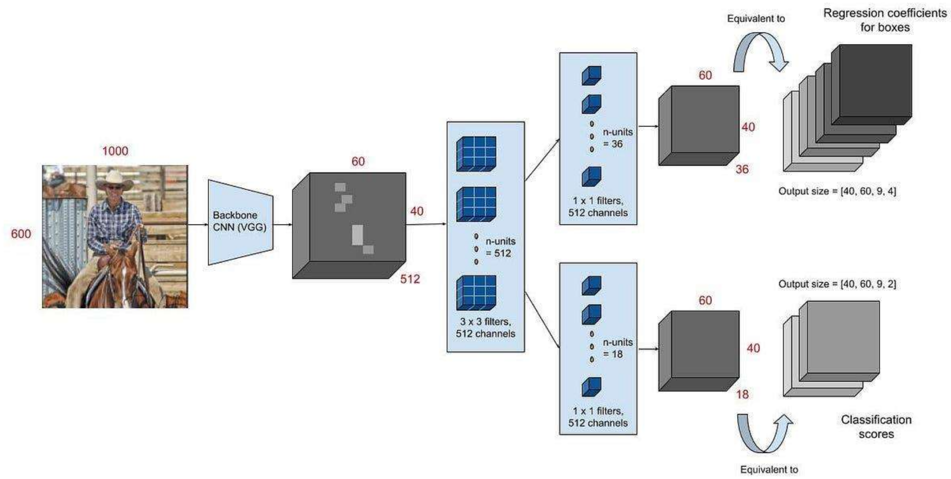

Figure 7. Faster R-CNN Architecture. Reprinted from [37].

Modern adaptations of region-based approaches incorporate attention mechanisms and feature pyramid networks to enhance multi-scale detection capabilities, enabling more effective identification of kidney abnormalities that vary substantially in size and appearance. These region-based approaches excel in medical imaging applications where detection accuracy takes precedence over processing speed, as the two-stage architecture allows for careful refinement of bounding box coordinates and classification decisions [38]. In urological imaging, region-based detectors demonstrate strength in identifying kidney abnormalities of varying sizes and morphologies, as the region proposal mechanism can adapt to detect both small focal lesions and larger masses within the same imaging study.

Recent implementations have integrated transformer-based feature extractors with region-based detection frameworks to capture long-range dependencies between anatomical structures, improving the discrimination of pathological findings from normal anatomical variations [39]. However, their computational requirements remain higher compared to single-shot alternatives, making them more suitable for offline analysis rather than real-time clinical applications.

2.2.4 Single-Shot and Real-Time Detection Architectures

Single-shot detection architectures emerged to address the computational limitations of region-based methods by reformulating object detection as a unified regression problem that simultaneously predicts bounding box coordinates and class probabilities. The You Only Look Once (YOLO) family of detectors has evolved through multiple iterations, with recent versions incorporating advanced backbone networks and sophisticated feature aggregation strategies that maintain real-time processing capabilities while substantially improving detection accuracy for medical imaging applications (see Figure 8 for a typical YOLO architecture) [40].

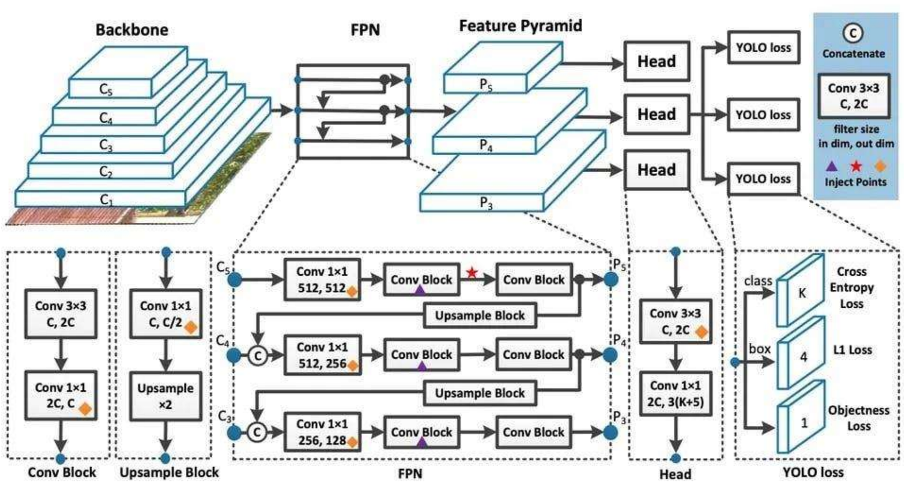

Figure **8**. *YOLO Architecture. Reprinted from [41].*

Contemporary YOLO applications in urological imaging have demonstrated remarkable success in kidney stone detection and kidney parenchyma localization. Recent

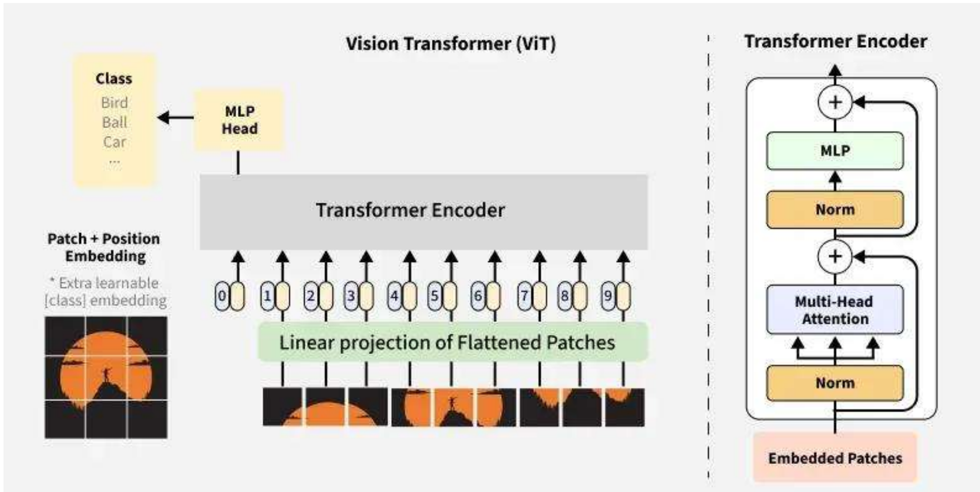

Figure 9. Vision Transformers. Reprinted from [44].

These transformer-based approaches demonstrate particular promise in medical imaging by leveraging attention mechanisms to focus computational resources on diagnostically relevant regions while maintaining awareness of broader anatomical context [38].

Recent applications in kidney imaging have explored dual-task approaches that combine detection with segmentation using transformer architectures. A transformer-based dual-task framework was presented for kidney MR segmentation in autosomal-dominant polycystic kidney disease, demonstrating the architecture's capacity to handle complex morphological variations while maintaining computational efficiency [38]. Transformers' showed superior ability to capture global spatial relationships essential for distinguishing pathological from normal tissue in challenging cases where local appearance alone is insufficient.

The ability of transformers to integrate information across entire imaging volumes proves advantageous for detecting kidney abnormalities that may be characterized by subtle texture changes or relationships to surrounding structures rather than dramatic

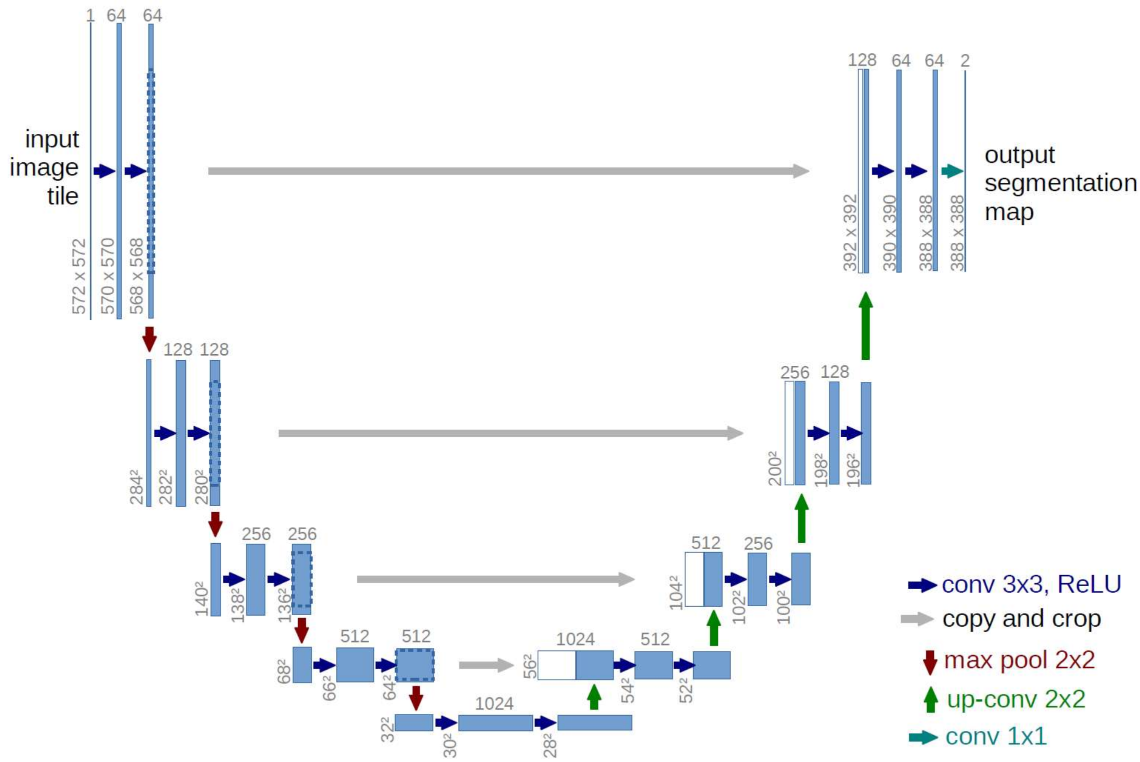

Figure 10. The U-Net Architecture. Illustration of the contracting and expanding paths with skip connections. Reprinted from [12].

The architecture consists of two main components defining its characteristic U-shaped structure. The contracting path captures context by progressively downsampling the input image, reducing spatial dimensions while increasing the number of feature channels to learn increasingly abstract representations [12]. Conversely, the symmetric expanding path enables precise localization by progressively upsampling feature maps. The critical innovation involves skip connections that directly link features from the contracting path to corresponding layers in the expanding path at the same resolution. These connections preserve fine-grained spatial details lost during downsampling, enabling the network to make precise predictions about pixel-level class assignments essential for accurate medical image segmentation [12].

Figure 11. Modality Versatility

This versatility stems from a modality-agnostic design that performs equally well on 2D image slices and volumetric 3D data after appropriate architectural modifications.

Clinical utility extends to numerous organ systems and pathological conditions, including brain tumor segmentation in multimodal MRI, cardiac structures in echocardiography, liver lesions in CT imaging, and lung structures in chest radiography [22], [46]. Recent comprehensive reviews of deep learning in medical image segmentation highlight U-Net's dominance across these applications, with the architecture serving as the baseline against which novel methods are evaluated [22]. This broad applicability reflects the robustness of the core architecture and its adaptability to task-specific modifications.

In kidney imaging specifically, U-Net has been extensively applied to diverse segmentation tasks. Studies have demonstrated its effectiveness for kidney parenchyma segmentation, renal mass delineation, and simultaneous segmentation of kidneys, tumors, and cysts in contrast-enhanced CT imaging [19], [47]. The computational efficiency of U-Net allows for the segmentation of high-resolution 3D volumes in seconds on contemporary GPU hardware, making integration into diagnostic workflows and surgical planning systems clinically feasible [12]. This rapid processing capability is particularly important for interventional procedures where real-time or near-real-time guidance is required.

2.3.4 U-Net Variants and Architectural Innovations

The success of the baseline architecture has prompted numerous refinements to handle increasingly complex segmentation tasks [22], [46]. One prominent variant, U-Net++, incorporates nested and dense skip pathways to bridge the semantic gap between feature maps extracted at different levels of the encoder and decoder (illustrated in Figure 12) [22].

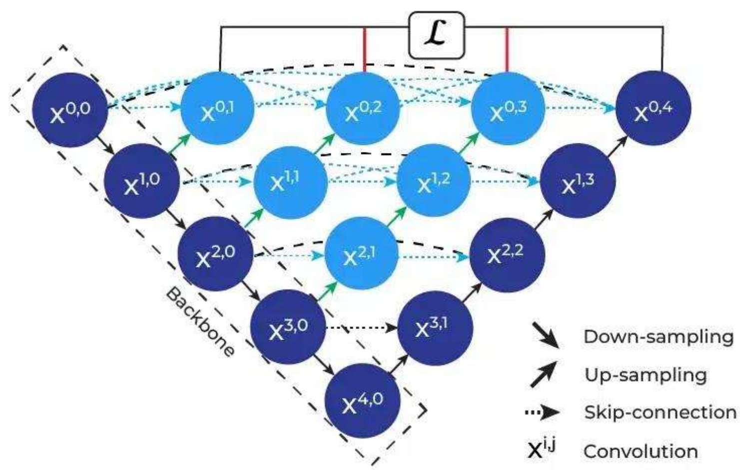

Figure 12. U-Net++ Architecture. Reprinted from [48].

By incorporating dense convolutional blocks at each skip connection node, U-Net++ reduces information loss during the decoding process and has demonstrated improved performance across multiple medical imaging datasets [22].

Attention-based variants have emerged as a major refinement direction, incorporating mechanisms to selectively focus on clinically relevant regions while suppressing background noise. Attention U-Net integrates attention gates within the skip connections to recalibrate feature maps based on spatial and channel-wise importance, improving segmentation accuracy particularly for small lesions (see the attention gate mechanism in Figure 13) [46].

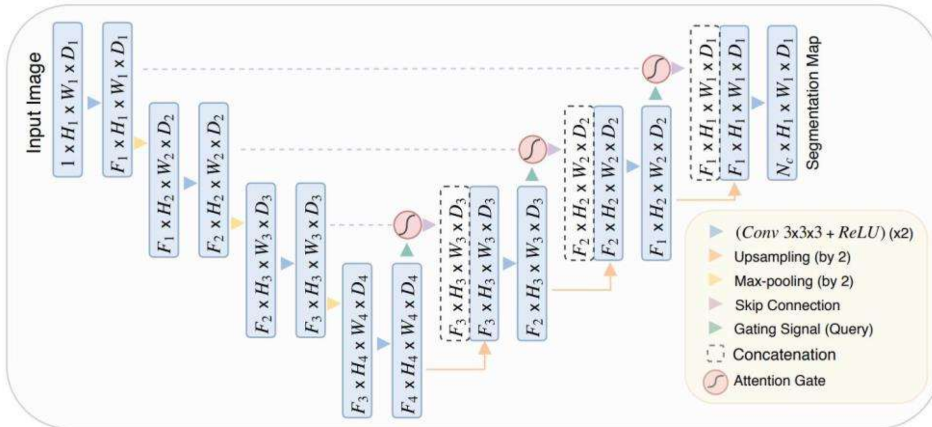

Figure 13. Attention Mechanism. Reprinted from [49].

Recent implementations specifically for kidney tumor segmentation have demonstrated that boundary attention mechanisms can significantly improve segmentation accuracy at tumor edges, addressing one of the primary challenges in renal mass characterization [50].

Multi-scale approaches represent another important variant class. MSS U-Net introduced multi-scale supervised learning for 3D kidney and tumor segmentation, training the network to produce accurate predictions at multiple resolution levels simultaneously [51]. This approach proved particularly effective for capturing both global kidney context and fine-grained tumor details. Similarly, recent work on the KiTS23 challenge explored various 3D U-Net training configurations including multi-scale prediction fusion, where separate models trained at different resolutions are combined through task-specific post-processing to achieve superior segmentation performance [36].

Furthermore, 3D U-Net extends the original architecture to volumetric medical imaging by replacing 2D convolutions with 3D convolutional operations, allowing direct processing of volumetric data [22], [46]. This extension has proven particularly valuable for kidney imaging where volumetric information is critical for accurate measurement and treatment planning. Recent studies employing 3D U-Net for renal mass segmentation have achieved Dice similarity coefficients exceeding 0.90 for kidney delineation and 0.75-0.83 for renal mass segmentation, demonstrating clinical-grade performance [17].

Hybrid variants have sought to combine U-Net's strengths with other architectural innovations. ResUNet combines the encoder-decoder structure with residual learning principles to leverage the representational power of residual connections [46]. More recently, transformer-based hybrid architectures have integrated self-attention mechanisms with the U-Net framework. TransUNet pioneered this approach by using a Vision Transformer as the encoder while maintaining U-Net's decoder structure (as depicted in Figure 14) [52].

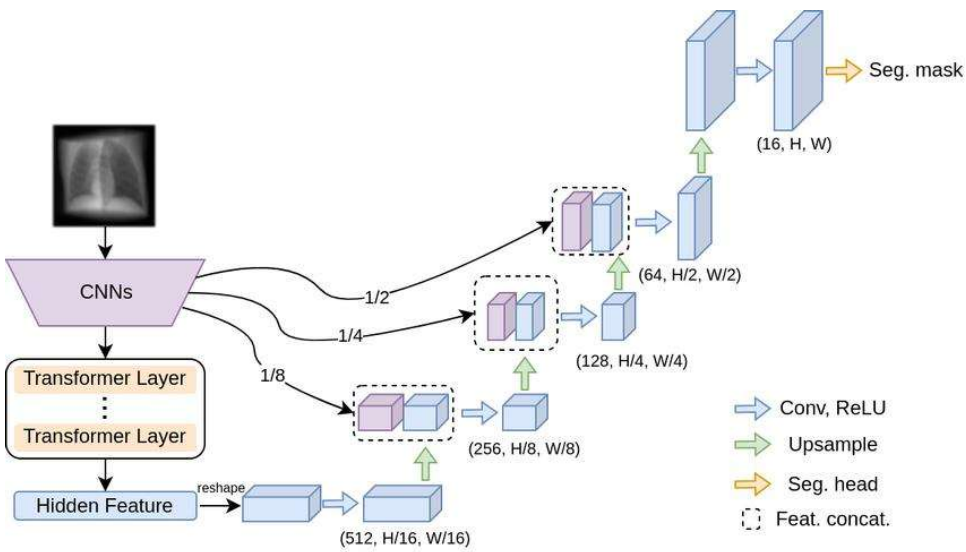

Figure 14. Hybrid Transformers. Reprinted from [53].

Swin-UNet further refined this concept using Swin Transformers with shifted window attention, demonstrating superior performance on multi-organ segmentation tasks including kidney structures [54].

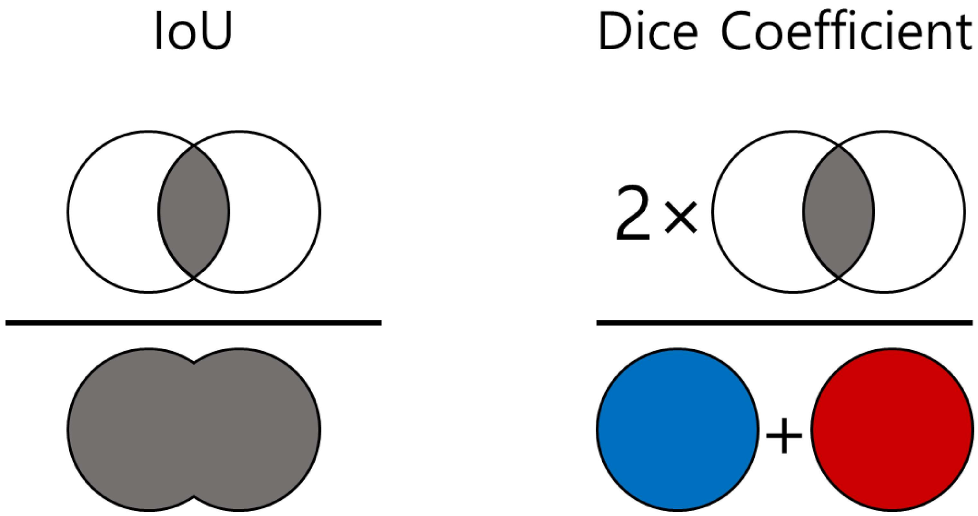

Figure 15. Volumetric Overlap Metrics

While these overlap-based metrics provide quantitative measures of volumetric accuracy, boundary precision is typically quantified using the Hausdorff distance and Average Symmetric Surface Distance [22].

Performance reporting typically includes sensitivity (recall), specificity, and precision to provide a comprehensive assessment of segmentation quality across different clinical requirements [22]. Sensitivity measures the proportion of actual positive pixels correctly identified, critical for ensuring complete lesion capture. Specificity quantifies the proportion of actual negative pixels correctly classified, important for minimizing false positive regions. Precision, or positive predictive value, indicates the proportion of predicted positive pixels that are actually positive, relevant for clinical confidence in segmented regions.

For clinical translation, published benchmarks suggest that Dice coefficients exceeding 0.85 for organ segmentation and 0.75 for lesion segmentation generally indicate clinically acceptable performance [22], [46]. However, mathematical accuracy metrics do not always correlate perfectly with clinical utility, necessitating a careful balance between metric optimization and the improvement of patient outcomes [22]. Recent studies on kidney and tumor segmentation have reported achieving these benchmarks, with state-of-the-art models attaining Dice scores of 0.948 for kidney segmentation, 0.776 for masses, and 0.738 for tumors on standardized challenge datasets [36].

Surface-based metrics have gained prominence for assessing boundary accuracy, particularly important in surgical planning where precise delineation of tumor margins is critical. The 95th percentile Hausdorff Distance (HD95) measures the maximum distance between predicted and ground truth boundaries while being robust to outliers, with lower values indicating better boundary alignment (illustrated in Figure 16).

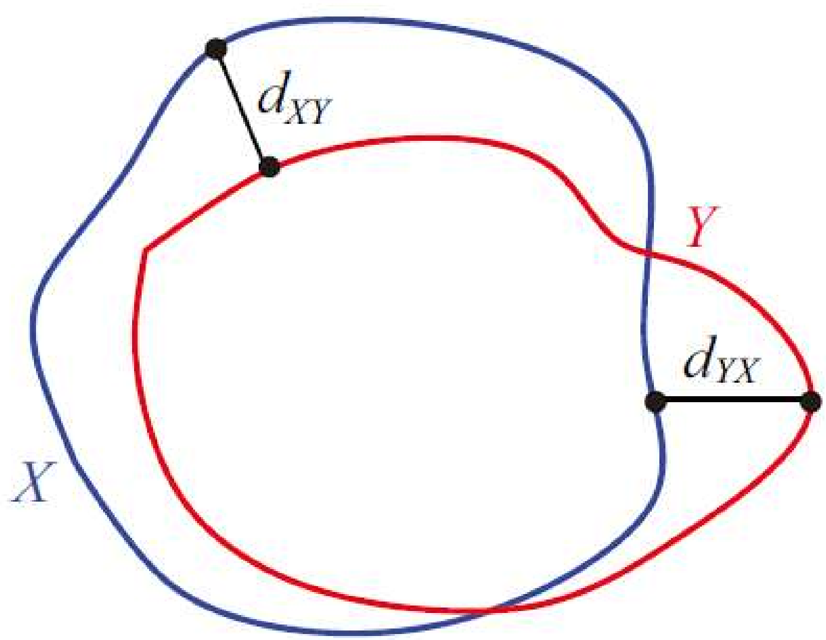

Figure 16. Boundary Distance Metrics. Reprinted from [57].

Recent kidney segmentation challenges have incorporated Surface Dice metrics that combine volumetric overlap with surface distance considerations, providing a more comprehensive assessment of segmentation quality for clinical applications [36].

The framework achieved first place in the KiTS19 kidney tumor segmentation challenge with a tumor Dice score of 0.85, establishing a benchmark for multi-institutional CT kidney segmentation [19]. Subsequent iterations have maintained dominance across biomedical segmentation tasks, with nnU-NetV2 demonstrating improved code structure, broader format support, and enhanced configurability while preserving the core automated configuration principles [20]. Recent comprehensive benchmarking against contemporary architectures including Vision Transformers and Mamba-based models has validated that properly configured CNN-based U-Net variants within the nnU-Net framework continue to achieve state-of-the-art performance when rigorously evaluated [21].

2.4.2 Automated Configuration Architecture

The framework implements automated configuration through a three-tier parameter hierarchy (shown in Figure 17) that systematically handles design decisions [20].

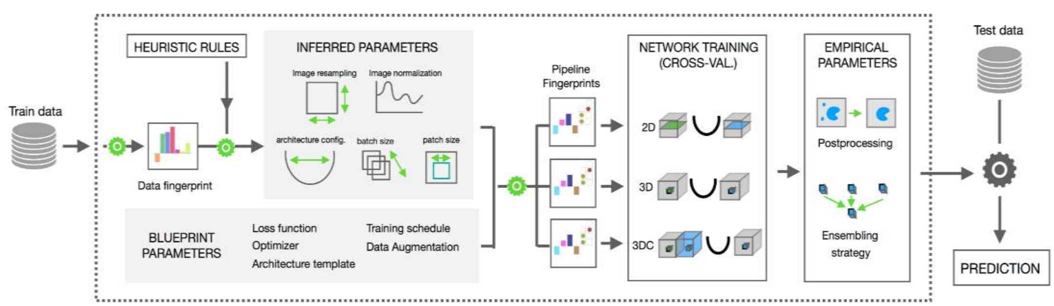

Figure 17. nnU-Net Configuration Hierarchy. Reprinted from [58]

The first tier consists of fixed parameters, which are design decisions that do not require adaptation between different datasets. These include the selection of a U-Net-like encoder-decoder architecture template, the choice of optimizer (SGD with Nesterov momentum), specific activation functions (leaky ReLU), and loss function strategies combining Dice and cross-entropy loss [20]. These fixed parameters emerged from

2.4.3 Cascade Architecture and Multi-Resolution Processing

A critical feature of the automated architecture selection is the systematic use of cascaded U-Net configurations for anatomical structures that occupy relatively small regions within imaging volumes [20]. The framework automatically detects when the configured patch size of the full-resolution 3D U-Net covers less than a specific threshold of the median image dimensions, triggering a two-stage cascade approach [20], [61]. In this configuration, the first U-Net operates on downsampled images to perform coarse localization of the target structures, while the second U-Net operates at full resolution within regions of interest identified by the first stage (see the cascade workflow in Figure 18) [20].

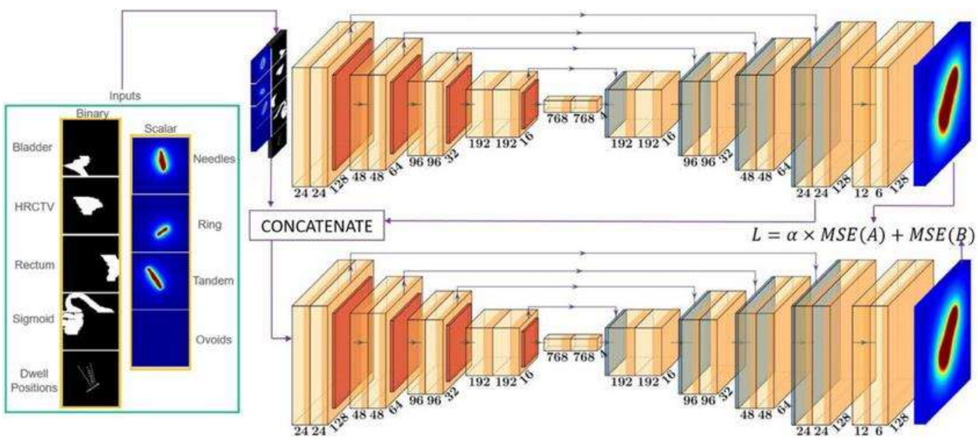

Figure 18. 3D Cascade Architecture. Retrieved from [62].

This cascade trigger directly addresses the fundamental trade-off between capturing sufficient contextual information for accurate localization and maintaining sufficient spatial resolution for precise boundary delineation [61], [63]. By automatically selecting cascade configurations based on the relationship between structure size and image dimensions, the system encodes domain knowledge about volumetric imaging without requiring manual intervention [20]. This design proves particularly valuable for kidney

2. Location: Ensuring representation of upper, mid, and lower pole lesions, as well as cortical vs. medullary placement.
3. Morphology: Selecting cases with varying densities (HU values) and shapes (e.g., simple vs. complex cysts, staghorn vs. focal stones).

This curation ensures that although the final dataset size is compact, it is mathematically representative of the disease spectrum, preventing the model from over-optimizing on "easy" or repetitive cases.

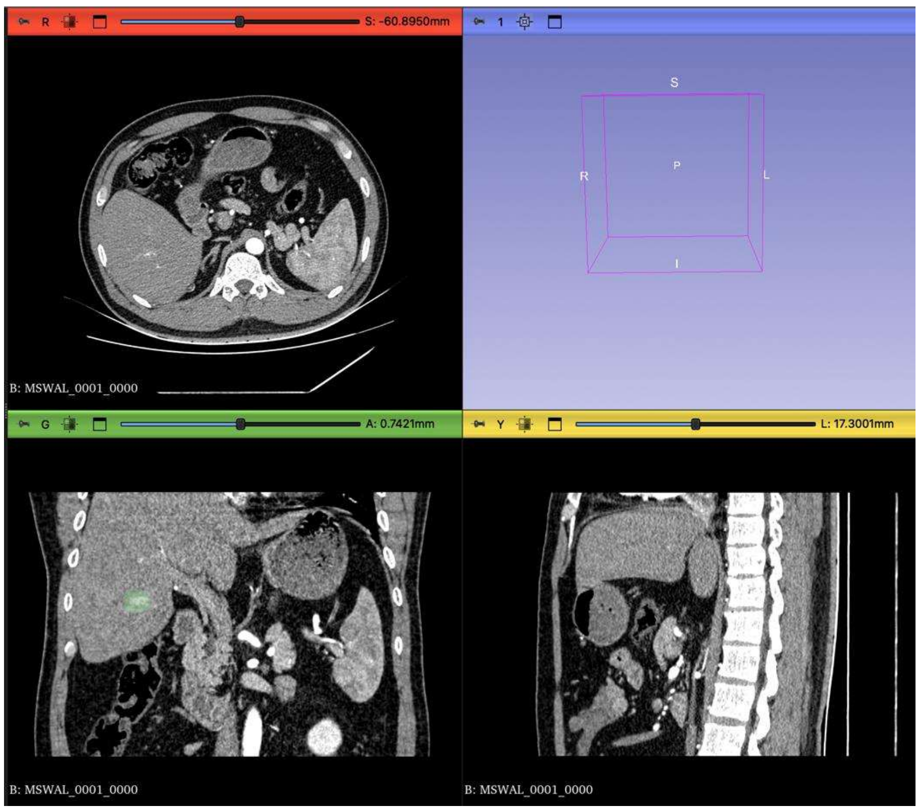

Figure 21. CT scan in different views

3.2.2 Ethical Compliance Framework:

1. Retrospective Study Protocol: Since this study utilizes publicly available, pre-anonymized datasets distributed under CC-BY-NC 4.0, institutional

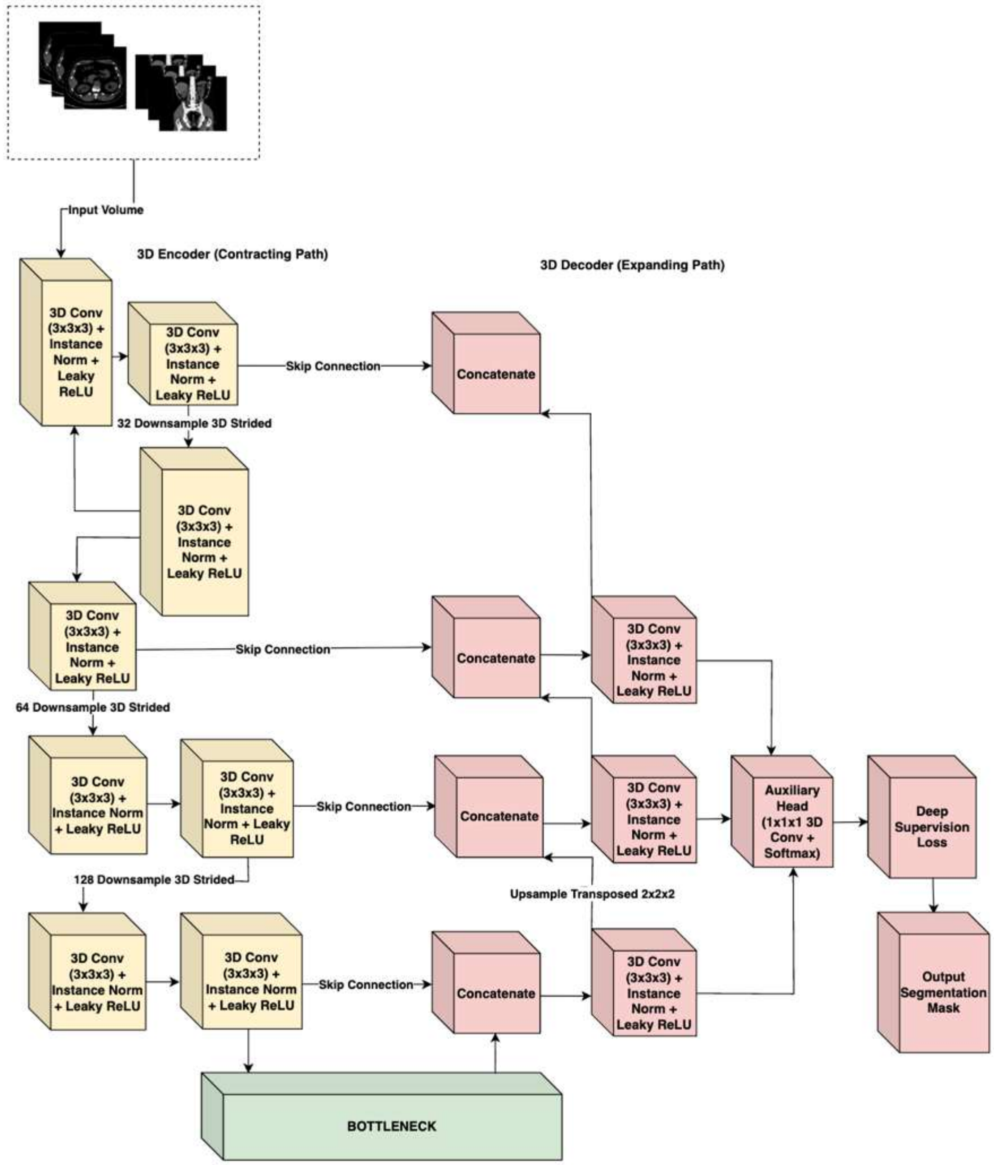

Figure 22. nnU-Net V2 Architecture

3.4.1 Architectural Topology

The network follows the classic symmetric Encoder-Decoder structure with skip connections, adapted for 3D inputs: